DANDY SLEEP EDUCATION

Sleep Apnea & SleepImage

A Quick-Start Guide for Dentists

A finger-based optical sensor records pulse-wave and oxygen-related changes during sleep.

Understand what the ring records, what the report is showing,
and which findings to review first.



The goal
You donotneed to become a sleep physician to understand the report. This guide gives you a practical reading order and plain-language definitions so you can confidently follow the patient's sleep-apnea journey.

Designed for dental teams beginning a dental sleep program

START HERE

Sleep Apnea: The Basics

The simplest definition
Sleepapnea is repeated interruption or reduction of breathing during sleep. An apnea is a pause in breathing; a hypopnea is a period of reduced or shallow breathing.

Two patterns shown on the report

Obstructive Sleep Apnea (OSA)	Central Sleep Apnea (CSA)
Breathing effort continues, but the upper airway is mechanically obstructed, interrupting airflow. SleepImage associates obstructive patterns with fragmentation and reports obstructive events separately.	There is a temporary lack of neural drive to breathe. SleepImage may identify periodic breathing patterns and reports central events separately.

Why the patient may feel unwell

Repeatedbreathingdisruptionscanloweroxygen, activate the body's stress response, interrupt sleep continuity and reducerestorativesleep.Apatientmayspendenoughtime in bedbut stillwake feeling unrefreshed.

Common complaints documented with the study

• Snoring or witnessed pauses in breathing	• Waking gasping or choking
• Excessive daytime sleepiness or fatigue	• Morning headaches
• Difficulty falling asleep or staying asleep	• Bruxism or teeth grinding

Important
Thereport supports clinical evaluation; it is not a stand-alone diagnosis. Results should be reviewed together with symptoms, medical history and the qualified clinician's interpretation.

THE TECHNOLOGY IN PLAIN LANGUAGE

What the SleepImage Ring Measures

Theringusesphotoplethysmography(PPG),a non-invasiveopticalmethodthatdetects changes in blood flow at the finger. The SleepImage software analyzes those signals during sleep.

1	**Pulse waveform and blood-volume changes:** Changes in the finger pulse signal provide information about circulation and breathing-related pressure changes.
2	**Heart rate and pulse-rate variability:** The timing between heartbeats changes with sleep depth, breathing andautonomicactivation.
3	**Oxygen saturation (SpO_2):** The ring tracks oxygen levels and identifies drops associated with respiratory events.
4	**Breathing-related changes:** Respiration alters the pulse waveform, allowing the system to evaluate breathing patterns without a nasal cannula or chest belt.
5	**Movement:** Actigraphy helps separate sleep, wake and REM-related patterns and helps identify motion-related signal disturbance.

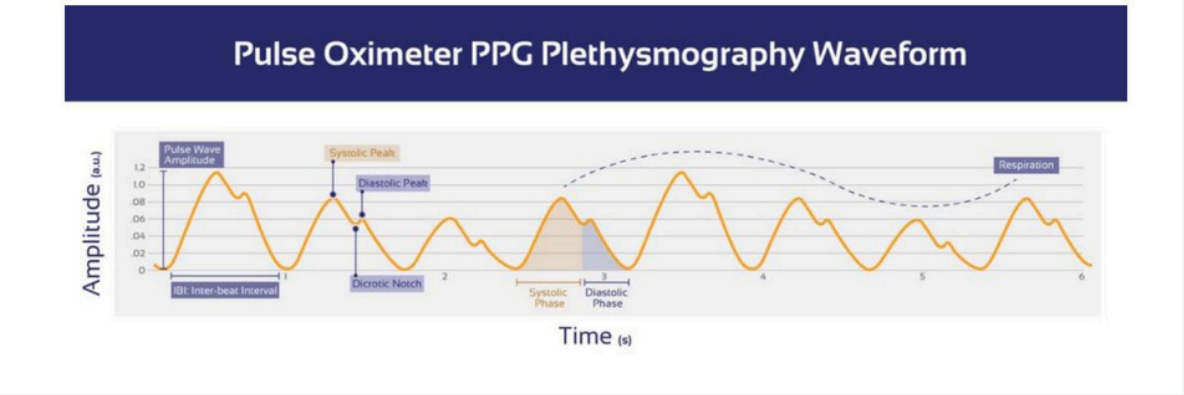


Figure from source page 19: respiration changes the finger pulse waveform, allowing breathing-related patterns to be evaluated.

How the software turns signals into a report

SleepImage uses cardiopulmonary coupling (CPC). In simple terms,the software looks at how pulse-rate variability and breathing activity move together over time. Those patterns areused to estimate sleep states, sleep stability and respiratory events.

What it is not

The ring does not directly record brainwaves with EEG. It evaluates sleep through autonomic, cardiovascular, respiratory, oxygen and movement signals collected at the finger.

USE THE SAME ORDER EVERY TIME

How to Read the Report in 60 Seconds

1	**Checksignalquality:**The signal line should bepredominantly green.Longyellow or red sections may represent noise or signal loss and can reduce confidence in those periods.
2	**Find the sAHI and severity:** The sAHI is the average number of apneas and hypopneas per hour of sleep. For adults: under 5 is below the sleep-apnea threshold; 5 to under 15 is mild; 15 to under 30 is moderate; 30 or more is severe.
3	**Confirm event type:** Review total, obstructive and central values. A report may contain one type or a mixtureofboth.
4	**Review oxygen impact:** Look at oxygen-desaturation index, minimum and mean SpO_2, time below 90%, 88% and 80%, and hypoxic burden.
5	**Compare sRDI with sAHI:** sRDI includes apnea/hypopnea events plus autonomic arousals that may disturb sleep without meeting the oxygen-desaturation criteria used by sAHI.
6	**Review sleep quality and stability:** Check SQI, sleep efficiency, stable sleep, unstable sleep, REM and wake transitions. These show how restorative and continuous the sleep was.
7	**Look for fragmentation and periodicity:** Fragmentation may be associated with obstruction or other causes of disrupted sleep. Periodicity may be associated with central or periodic breathing patterns.
8	**Read the automated summary:** Use it as a quick cross-check, then confirm the numbers and clinical context rather than relying on the summary alone.

Dentistshortcut

Start with signal quality, sAHI, obstructive versus central events, oxygen and the physician’s impression. The remaining metrics help explain how the breathing disorder affected the patient’s sleep.

REPORT TERMINOLOGY

The Breathing Numbers That Matter

Metric	Plain-English meaning	What to notice
sAHI 3% and 4%	Average respiratory events per hour using the report's 3% and 4% oxygen-desaturation scoring views.	Severity category and whether events are obstructive or central.
sRDI	sAHI plus additional autonomic arousals that may disturb sleep without a qualifying oxygen drop.	A higher sRDI than sAHI may suggest additional breathing-related sleep disruption.
ODI	Number of oxygen-desaturation events per hour.	How frequently oxygen falls during the night.
SpO_2 summary	Minimum, maximum and mean oxygen, plus time below 90%, 88% and 80%.	Depth and duration of low-oxygen exposure, not only the single lowest value.
Hypoxic Burden	Combines the frequency, depth and duration of respiratory-event-related oxygen drops.	A broader view of oxygen stress than AHI alone.
SAI / CVHR	Cardiac response pattern associated with repeated breathing and oxygen changes.	Provides added context about how respiratory events affect sleep physiology.

Adult sAHI severity categories

Below threshold	Mild	Moderate	Severe
< 5	5 to < 15	15 to < 30	≥ 30

Pediatric reports use different thresholds
Forchildren,thereportuseslowersAHI/sRDI/SAImarkers: mild ≥1, moderate ≥5 and severe ≥10 events per hour.

Do not stop at AHI
AHIcounts eventfrequency, but it does not fully describe event duration, oxygen depth, sleep fragmentation or symptoms. Review the oxygen, sRDI, sleep-quality and pathology sections as well.

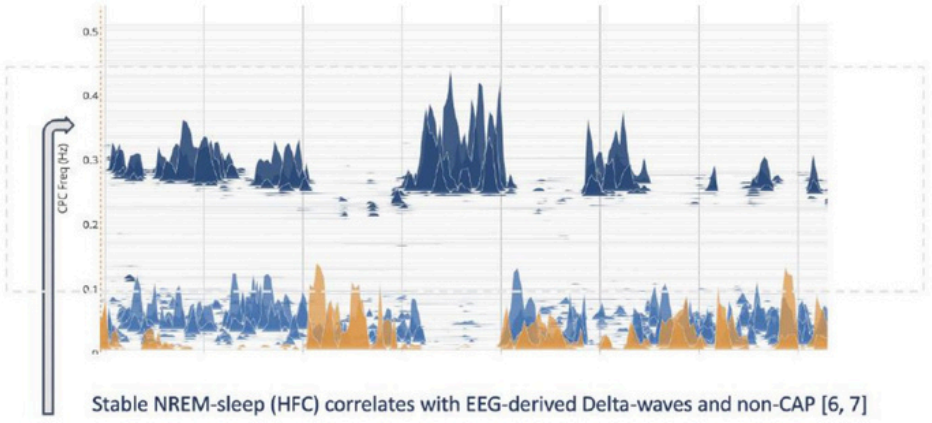
WHY THE PATIENT CAN SLEEP “ALL NIGHT” AND STILL FEEL TIRED

Understanding Sleep Quality and Stability

Metric	What it means	Expected / helpful reference
SQI	A 0–100 summary of sleep quality, stability, fragmentation and periodicity.	Adults >55; children >70
Sleep Efficiency	Percentage of time allocated to sleep that was actually spent asleep.	Generally >85%
Sleep Latency	Time from the beginning of the sleep opportunity to sleep onset.	Generally <30 minutes
Stable Sleep	Restorative NREM sleep associated with stable breathing, oxygenation and autonomic regulation.	Adults >50%; children >65%
Unstable Sleep	NREM sleep with variable breathing, heart-rate patterns and arousal thresholds.	Interpret with fragmentation, symptoms and other metrics
Fragmentation (e-LFCBB)	Marker of disrupted sleep; may occur with OSA, upper-airway resistance, pain, insomnia or movement disorders.	Adults <15; children <8
Periodicity (e-LFCNB)	Marker of periodic-type breathing patterns that may occur with central apnea or periodic breathing.	Adults ≤2; children 0

A few interpretation rules

- A good sleepefficiency does notrule out sleep apnea. A patient may fall asleep easily and remain asleep while still having repeated respiratory events.
- A low SQI means the overall sleep pattern was less healthy or stable, but the cause must be interpreted from the rest of the report and clinical history.
- Fragmentation is sensitive but not specific. It can be increased by obstruction, insomnia, pain, anxiety or movement disorders.
- Periodicity should be reviewed together with central events, oxygen patterns and the interpreting clinician’s assessment.
- One night is a snapshot. Multi-night testing may provide useful context when symptoms and findings do not align.

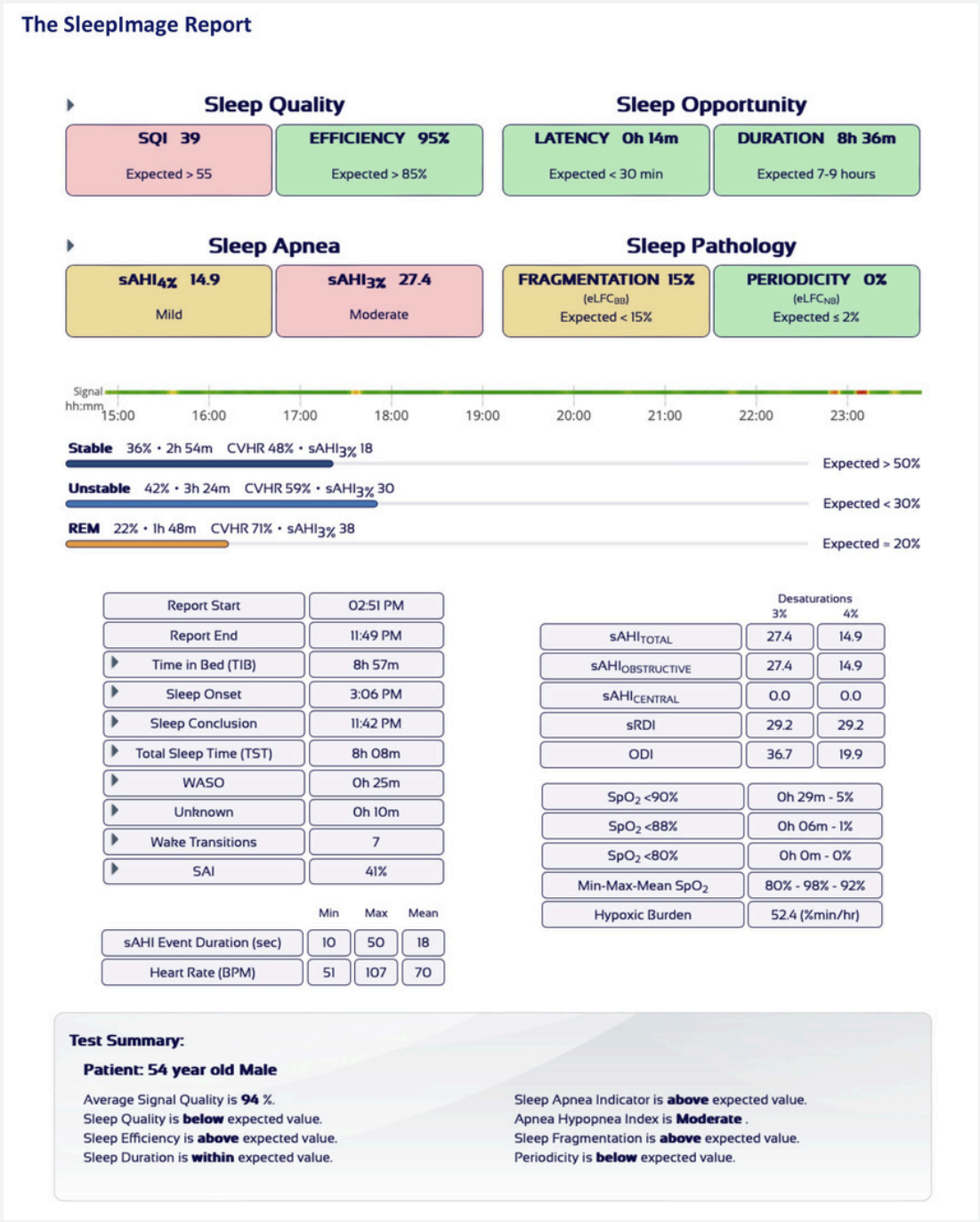


Example from source page 12: stronger upper peaks represent stable NREM sleep; lower-frequency activity reflects less stable sleep patterns.

Key message
The best report review combines three questions: How often did breathing events occur? How much did oxygen fall? How much did those events disturb restorative sleep?

APPLY THE READING ORDER

Example Report: A Guided Walk-Through



The SleepImage Report

Sleep Quality

SQI 39	EFFICIENCY 95%
Expected > 55	Expected > 85%

Sleep Opportunity

LATENCY 0h 14m	DURATION 8h 36m
Expected < 30 min	Expected 7-9 hours

Sleep Apnea

$sAHI_{4\%}$ 14.9	$sAHI_{3\%}$ 27.4
Mild	Moderate

Sleep Pathology

FRAGMENTATION 15% ($eLFC_{BB}$)	PERIODICITY 0% ($eLFC_{NB}$)
Expected < 15%	Expected ≤ 2%

Signal hh:mm 15:00 16:00 17:00 18:00 19:00 20:00 21:00 22:00 23:00

Stable 36% • 2h 54m CVHR 48% • $sAHI_{3\%}$ 18 — Expected > 50%

Unstable 42% • 3h 24m CVHR 59% • $sAHI_{3\%}$ 30 — Expected < 30%

REM 22% • 1h 48m CVHR 71% • $sAHI_{3\%}$ 38 — Expected ≈ 20%

Report Start	02:51 PM
Report End	11:49 PM
Time in Bed (TIB)	8h 57m
Sleep Onset	3:06 PM
Sleep Conclusion	11:42 PM
Total Sleep Time (TST)	8h 08m
WASO	0h 25m
Unknown	0h 10m
Wake Transitions	7
SAI	41%

	Min	Max	Mean
sAHI Event Duration (sec)	10	50	18
Heart Rate (BPM)	51	107	70

	Desaturations 3%	4%
$sAHI_{TOTAL}$	27.4	14.9
$sAHI_{OBSTRUCTIVE}$	27.4	14.9
$sAHI_{CENTRAL}$	0.0	0.0
sRDI	29.2	29.2
ODI	36.7	19.9

SpO_2 <90%	0h 29m - 5%
SpO_2 <88%	0h 06m - 1%
SpO_2 <80%	0h 0m - 0%
Min-Max-Mean SpO_2	80% - 98% - 92%
Hypoxic Burden	52.4 (%min/hr)

Test Summary:

Patient: 54 year old Male

Average Signal Quality is **94** %.
Sleep Quality is **below** expected value.
Sleep Efficiency is **above** expected value.
Sleep Duration is **within** expected value.

Sleep Apnea Indicator is **above** expected value.
Apnea Hypopnea Index is **Moderate** .
Sleep Fragmentation is **above** expected value.
Periodicity is **below** expected value.

Complete sample report reproduced from source page 33. The next page walks through the findings in plain language.

What this sample is showing

Area	Sample finding	Plain-language takeaway
Signal	Average signal quality 94%	The recording is predominantly usable.
Sleep quality	SQI 39; stable sleep 36%	Overall sleep quality and restorative stable sleep are below expected adult values.
Sleep opportunity	Efficiency 95%; duration 8 h 36 min	The patient had adequate sleep opportunity and spent most of it asleep.
Sleep apnea	sAHI 4% = 14.9; sAHI 3% = 27.4	The two scoring views fall in mild and moderate adult categories.
Event type	Obstructive values match total; central = 0	The respiratory events in this sample are predominantly obstructive.
Additional disturbance	sRDI 29.2; fragmentation 15%	Breathing-related arousals and sleep disruption deserve review alongside sAHI.
Oxygen	Mean 92%; minimum 80%; 5% below 90%; HB 52.4	The oxygen section adds important severity context beyond event count.
Periodicity	0%	The sample does not show an elevated periodicity marker.

Plain-English summary of the sample

Thisisa good-quality recording withefficient sleep, but the report shows predominantly obstructive respiratory events, reduced sleep quality and stable sleep, and oxygen findings that require clinical interpretation. The final diagnosis and treatment recommendation should come from the qualified interpreting clinician.

QUALITY CHECKS AND CLINICAL CONTEXT

When the Report Needs a Closer Look

1	**Longredsignalsections:**Signallossornoisecanmakethoseperiodsunreliable. A repeat study may be appropriate when signal loss is prolonged.
2	**Central events or elevated periodicity:** These findings require careful clinical review rather than assuming thepatternisobstructive.
3	**Low oxygen or prolonged time below thresholds:** Review minimum and mean SpO_2, time below 90%/88%/80% and hypoxic burden together.
4	**sRDI notably higher than sAHI:** Additional autonomic arousals may be disrupting sleep even when they do notmeetsAHIoxygencriteria.
5	**Symptoms do not match the numbers:** Consider the full history, sleep quality, signal quality and whether multi-night data may provide a clearer picture.
6	**Good efficiency but low SQI:** The patient may sleep for a long time without getting stable, restorative sleep.

Mini glossary

PPG: Optical finger measurement of blood-flow and pulse-wave changes.	**CPC:** Cardiopulmonary coupling: analysis of the relationship between pulse-rate variability and breathing.
sAHI: SleepImage Apnea Hypopnea Index: apneas and hypopneas per hour of sleep.	**sRDI:** Respiratory disturbance index including additional autonomic arousals.
ODI: Oxygen desaturation events per hour.	**SpO_2:** Blood oxygen saturation.
SQI: Sleep Quality Index on a 0–100 scale.	**OSA / CSA:** Obstructive sleep apnea / central sleep apnea.

Clinical reminder

Expected values are guides, not absolute diagnostic cutoffs. The SleepImage report should be interpreted with the patient's symptoms, medical history, medications and the responsible clinician's evaluation.

Source

Adaptedand simplified from Introduction to SleepImage, Revision 34, published July 10, 2026. Selected figures are reproduced from pages 12, 18, 19 and 33 for clinician education. Always refer to the current SleepImage Instructions for Use and full clinician guide for complete technical and regulatory information.